$$P(\text{choosing an even integer}) = P(A) = \frac{n(A)}{n(S)} = \frac{5}{10} = \frac{1}{2}.$$

$$P(\text{choosing an integer multiple of three}) = P(B) = \frac{n(B)}{n(S)} = \frac{3}{10}.$$

Example 12.3

Three coins are tossed simultaneously, what is the probability of getting (i) exactly one head (ii) at least one head (iii) at most one head?

Solution:

Notice that three coins are tossed simultaneously = one coin is tossed three times.

The sample space $S = \{H,T\} \times \{H,T\} \times \{H,T\}$

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\},\ n(S) = 8$$

Let A be the event of getting exactly one head, B be the event of getting atleast one head and C be the event of getting at most one head.

$$A = \{HTT, THT, TTH\};\ n(A) = 3$$

$$B = \{HTT, THT, TTH, HHT, HTH, THH, HHH\};\ n(B) = 7$$

$$C = \{TTT, HTT, THT, TTH\};\ n(C) = 4.$$

Therefore the required probabilities are

(i) $P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$

(ii) $P(B) = \frac{n(B)}{n(S)} = \frac{7}{8}$

(iii) $P(C) = \frac{n(C)}{n(S)} = \frac{4}{8} = \frac{1}{2}.$

Note 12.3

When the number of elements in a sample space is considerably small we can solve by finger-counting the elements in the events. But when the number of elements is too large to count then combinatorics helps us to solve the problems.



For the following problem, combinatorics is used to find the number of elements in the sample space and the events.

Example 12.4

Suppose ten coins are tossed. Find the probability to get (i) exactly two heads (ii) at most two heads (iii) at least two heads

Solution

Ten coins are tossed simultaneously one time = one coin is tossed 10 times

Let S the sample space,

That is $$S = \overbrace{\{H,T\} \times \{H,T\} \times \{H,T\} \times \cdots \times \{H,T\}}^{10 \text{ times}}$$

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